

GCSE Mathematics

Predicted Paper 3 — Higher Tier

Set B

Pearson Edexcel style (1MA1 / 3H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

This is an original predicted paper produced by GCSEMathsAI for revision purposes only. It is not affiliated with or endorsed by Pearson Edexcel.

Answer ALL questions. Write your answers in the spaces provided.

1. A coat normally costs £85. In a sale, the price is reduced by 35%. Three weeks later, the sale price is reduced by a further 20%. Work out the final sale price.

(3 marks)

2. Work out $(4.5 \times 10^7) \div (1.5 \times 10^2) + 2 \times 10^5$. Give your answer in standard form.

(3 marks)

3. Jenny and Kate share a sum of money in the ratio 5 : 7. Kate gives £36 to Jenny so that they each have an equal amount. Work out the original sum of money they shared.

(4 marks)

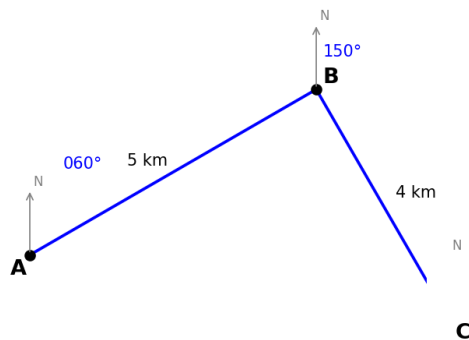
4. Solve $(x + 3)/4 + (2x - 1)/3 = 5$. Show your working clearly.

(3 marks)

5. Asif invests £2,500 in a savings account paying compound interest at a fixed rate per year. After 2 years, the value of his investment is £2,652.25. Work out the annual interest rate.

(4 marks)

6. A boat sails from port A on a bearing of 060° for 5 km to point B. From B, it sails on a bearing of 150° for 4 km to point C.



(a) Show that angle $ABC = 90^\circ$.

(2 marks)

(b) Calculate the direct distance from A to C. Give your answer correct to 3 s.f.

(2 marks)

(c) Work out the bearing of C from A. Give your answer to the nearest degree.

(3 marks)

7. Simplify each expression. Give your answer in its simplest form.

(a) $\sqrt{72} + \sqrt{50}$

(2 marks)

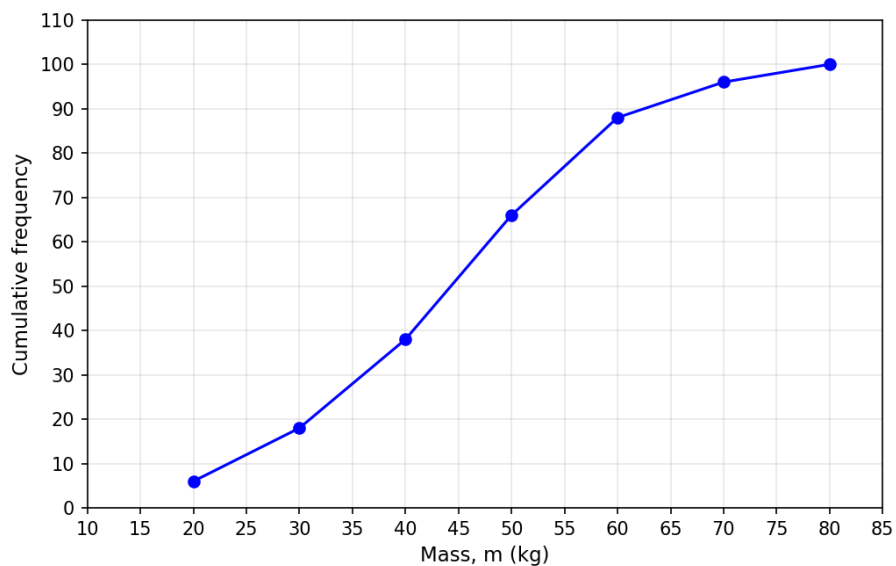
(b) $8 \div \sqrt{2}$

(2 marks)

8. It takes 4 builders 12 days to build a wall. Assuming all builders work at the same rate, how many days would it take 6 builders to build the same wall? Show your working.

(3 marks)

9. The cumulative frequency curve shows the masses of 100 packages.



(a) Find an estimate for the median mass.

(1 marks)

(b) Find an estimate for the interquartile range.

(2 marks)

(c) Estimate the number of packages with mass greater than 65 kg.

(2 marks)

10. By completing the square, solve $x^2 - 6x + 2 = 0$. Give your answers in the form $a \pm b\sqrt{c}$, where a , b , c are integers.

(4 marks)

11. A drawer contains 5 black socks and 4 white socks. Two socks are taken at random from the drawer without replacement.

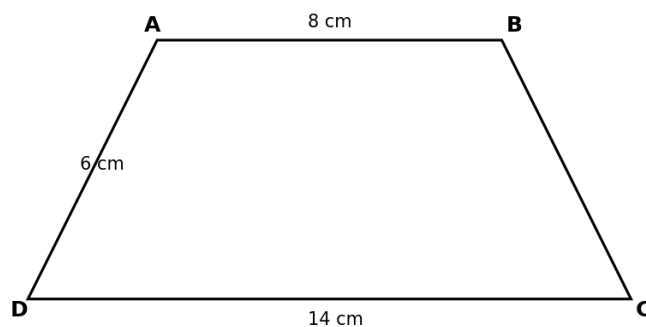
(a) Work out the probability that both socks are the same colour.

(3 marks)

(b) Given that the two socks are the same colour, find the probability that they are both black.

(3 marks)

12. The diagram shows trapezium ABCD. AB is parallel to DC. AB = 8 cm, DC = 14 cm, AD = 6 cm, and angle ADC = 90° .



(a) Work out the area of trapezium ABCD.

(2 marks)

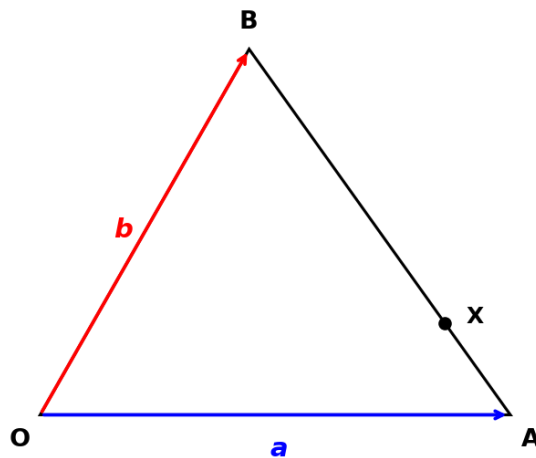
(b) Work out the length of BC. Give your answer correct to 3 significant figures.

(3 marks)

13. A line segment has endpoints (1, 4) and (5, -2). Find the equation of the perpendicular bisector of this line segment. Give your answer in the form $y = mx + c$.

(4 marks)

14. OAB is a triangle. $\mathbf{OA} = \mathbf{a}$ and $\mathbf{OB} = \mathbf{b}$. X is the point on AB such that $AX : XB = 1 : 3$.



(a) Express $\mathbf{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$. Give your answer in its simplest form.

(3 marks)

(b) Y is the point on OB such that $OY = \frac{1}{4}OB$. Show that XY is parallel to OA.

(3 marks)

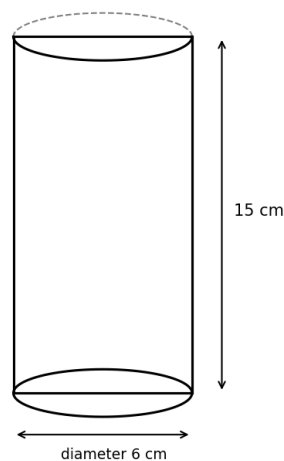
15. Show that $(n + 2)^2 - (n - 2)^2$ is always a multiple of 8 for any integer n .

(3 marks)

16. Make x the subject of the formula $y = (3x + 1) / (2 - x)$.

(4 marks)

17. A solid cylinder has a diameter of 6 cm and a height of 15 cm.



(a) Work out the volume of the cylinder. Give your answer in terms of π .

(2 marks)

(b) Work out the total surface area of the cylinder (including the top and bottom). Give your answer correct to 3 significant figures.

(4 marks)

18. The mass M , in grams, of a radioactive sample after t years is given by $M = 200 \times (0.94)^t$.

(a) Find the initial mass of the sample (when $t = 0$).

(1 marks)

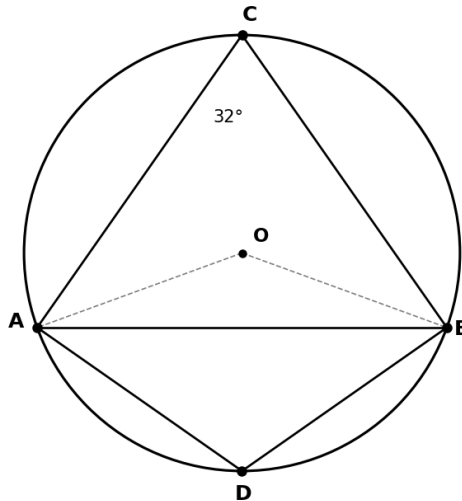
(b) Find the mass after 10 years. Give your answer to 1 decimal place.

(2 marks)

(c) By trial and improvement (or otherwise), find the time taken for the mass to fall to half its initial value. Give your answer to the nearest year.

(3 marks)

19. A, B, C and D are points on the circumference of a circle, centre O. Angle $ACB = 32^\circ$.



(a) Work out the size of angle AOB (reflex angle through the major arc is not needed; give the angle at the centre standing on the same arc as ACB). Give a reason.

(2 marks)

(b) Work out the size of angle ADB. Give a reason for your answer.

(2 marks)

20. A solution of the equation $x^3 + 2x - 7 = 0$ lies between 1 and 2.

(a) Show that the equation $x^3 + 2x - 7 = 0$ can be rearranged to $x = (7 - 2x)^{1/3}$ (the cube root of $7 - 2x$).

(2 marks)

(b) Starting with $x_0 = 1.5$, use the iteration formula $x_{n+1} = (7 - 2x_n)^{1/3}$ to find x_1 , x_2 and x_3 . Give each value correct to 4 decimal places.

(3 marks)

21. The table shows the number of goals scored in each of 30 football matches.

Goals scored	0	1	2	3	4	5
Frequency	5	8	7	x	3	2

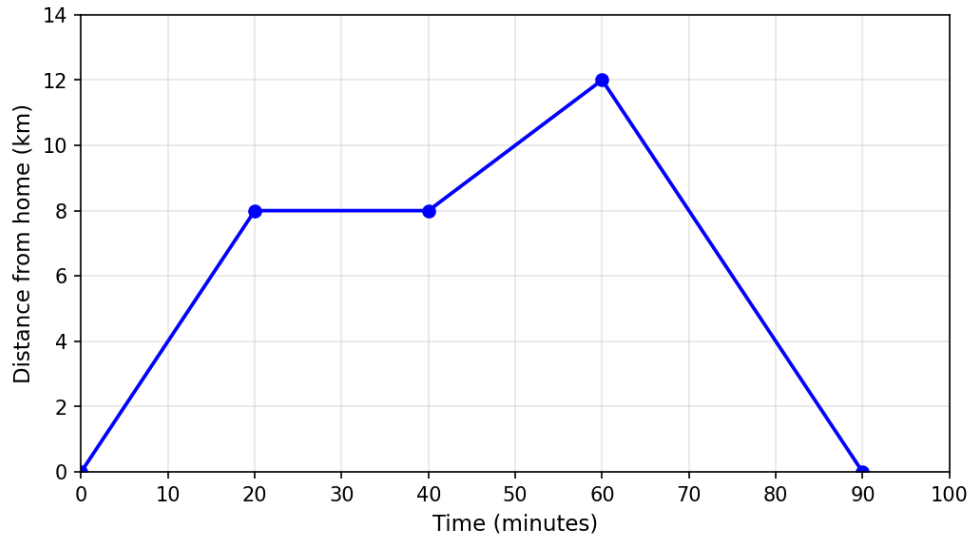
(a) Find the value of x.

(2 marks)

(b) Calculate the mean number of goals scored per match. Give your answer to 2 d.p.

(3 marks)

22. The graph shows Jamie's cycling journey from home and back.



(a) How long did Jamie stop for during the journey?

(1 marks)

(b) Calculate the average speed for the entire return journey (from leaving home to returning). Give your answer in km/h.

(3 marks)

(c) During which section of the journey was Jamie travelling fastest? Justify your answer with calculation.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 3 Set B

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	£44.20	M1: $85 \times 0.65 = 55.25$ (after first reduction). M1: 55.25×0.80 . A1: 44.20. [3]
2	5×10^5	M1: $4.5 \div 1.5 = 3$ AND $10^7 \div 10^2 = 10^5$. M1: $3 \times 10^5 + 2 \times 10^5$. A1: 5×10^5 . [3]
3	£432	M1: $5 + 7 = 12$ parts; for equal split, each gets 6 parts. M1: Kate gives 1 part to Jenny so they end up $6 : 6$. So 1 part = 36. M1: identifies 12 parts total. A1: $12 \times 36 = 432$. [4]
4	$x = 51/11$	M1: multiplies through by 12: $3(x + 3) + 4(2x - 1) = 60$. M1: $3x + 9 + 8x - 4 = 60 \Rightarrow 11x + 5 = 60$. A1: $x = 55/11 = 5$. <i>Check arithmetic: $11x = 55 \Rightarrow x = 5$.</i> [3]
5	3% per year	M1: $2500 \times (1 + r)^2 = 2652.25$. M1: $(1 + r)^2 = 1.0609$. M1: $1 + r = \sqrt{1.0609} = 1.03$. A1: $r = 0.03 = 3\%$. [4]
6(a)	Shown: angle ABC = 90°	M1: the change in bearing from 060° (extended) to 150° is the interior angle at B. The reverse direction of AB has bearing $060 + 180 = 240$. New bearing BC = 150 . Interior angle = $240 - 150 = 90^\circ$. A1: clear explanation. [2]
6(b)	AC = 6.40 km (3 s.f.)	M1: Pythagoras $AC^2 = 5^2 + 4^2 = 41$. A1: $AC = \sqrt{41} = 6.40$ km. [2]
6(c)	Bearing $\approx 098^\circ$	M1: angle BAC by trig: $\tan(\text{BAC}) = 4/5 \Rightarrow \text{angle BAC} = 38.66^\circ$. M1: bearing of C from A = $060 + 38.66 = 98.66^\circ$. A1: 099° (nearest degree). (<i>Accept 098 or 099.</i>) [3]
7(a)	$11\sqrt{2}$	M1: $\sqrt{72} = 6\sqrt{2}$ AND $\sqrt{50} = 5\sqrt{2}$. A1: $6\sqrt{2} + 5\sqrt{2} = 11\sqrt{2}$. [2]
7(b)	$4\sqrt{2}$	M1: $8\sqrt{2} \times \sqrt{2}/\sqrt{2} = 8\sqrt{2}/2$. A1: $4\sqrt{2}$. [2]
8	8 days	M1: identifies inverse proportion: total builder-days = $4 \times 12 = 48$. M1: $48 / 6$. A1: 8 days. [3]
9(a)	Median ≈ 47 kg	B1: reads at cumulative frequency 50: $t \approx 47$ (accept 45–48). [1]
9(b)	IQR ≈ 18 kg	M1: identifies Q1 at cf = 25 (≈ 35) and Q3 at cf = 75 (≈ 53). A1: IQR ≈ 18 (accept 15–20). [2]
9(c)	≈ 16 packages	M1: at $m = 65$, cf ≈ 84 . M1: $100 - 84 = 16$. A1: 16 (accept 12–20). [2]
10	$x = 3 \pm \sqrt{7}$	M1: $(x - 3)^2 - 9 + 2 = 0$. M1: $(x - 3)^2 = 7$. M1: $x - 3 = \pm\sqrt{7}$. A1: $x = 3 \pm \sqrt{7}$. [4]
11(a)	$16/36 = 4/9$	M1: $P(\text{BB}) + P(\text{WW}) = (5/9)(4/8) + (4/9)(3/8)$. M1: $= 20/72 + 12/72 = 32/72$. A1: $4/9$. [3]
11(b)	$5/8$	M1: identifies conditional probability = $P(\text{BB}) / [P(\text{BB}) + P(\text{WW})] = (20/72) / (32/72)$. A1: $20/32 = 5/8$. [3]
12(a)	66 cm^2	M1: $(1/2)(8 + 14) \times 6$. A1: 66. [2]
12(b)	BC ≈ 8.49 cm (3 s.f.)	M1: drops a perpendicular from B to DC, meeting at E, where $DE = AB = 8$. Therefore $EC = 14 - 8 = 6$ and $BE = AD = 6$. M1: Pythagoras $BC^2 = 6^2 + 6^2 = 72$. A1: $BC = \sqrt{72} = 6\sqrt{2} = 8.49$ cm (3 s.f.). [3]
13	$y = (2/3)x - 1$	M1: midpoint = (3, 1). M1: original gradient = $(-2 - 4)/(5 - 1) = -3/2$. M1: perpendicular gradient = $2/3$. M1: $y - 1 = (2/3)(x - 3)$. A1: $y = (2/3)x - 1$. [4]
14(a)	$\mathbf{OX} = (3/4)\mathbf{a} + (1/4)\mathbf{b}$	M1: $\mathbf{OX} = \mathbf{OA} + (1/4)\mathbf{AB}$. M1: $\mathbf{AB} = \mathbf{b} - \mathbf{a}$, so $\mathbf{OX} = \mathbf{a} + (1/4)(\mathbf{b} - \mathbf{a})$. A1: $= (3/4)\mathbf{a} + (1/4)\mathbf{b}$. [3]

Q	Answer	Mark scheme
14(b)	Shown	M1: $\mathbf{OY} = (1/4)\mathbf{b}$. $\mathbf{XY} = \mathbf{OY} - \mathbf{OX} = (1/4)\mathbf{b} - (3/4)\mathbf{a} - (1/4)\mathbf{b} = -(3/4)\mathbf{a}$. M1: clearly states $\mathbf{XY}$ is a scalar multiple of $\mathbf{a}$ ($= \mathbf{OA}$). A1: concludes XY is parallel to OA. [3]
15	Shown	M1: $(n+2)^2 - (n-2)^2 = (n^2 + 4n + 4) - (n^2 - 4n + 4)$. M1: simplifies $= 8n$. A1: clearly states $8n$ is a multiple of 8 for any integer n . [3]
16	$x = (2y - 1) / (y + 3)$	M1: $y(2 - x) = 3x + 1$. M1: $2y - xy = 3x + 1$. M1: $2y - 1 = 3x + xy = x(3 + y)$. A1: $x = (2y - 1) / (y + 3)$. [4]
17(a)	$135\pi \text{ cm}^3$	M1: $V = \pi r^2 h = \pi(3)^2(15)$. A1: 135π . [2]
17(b)	339 cm^2 (3 s.f.)	M1: curved surface area $= 2\pi rh = 2\pi(3)(15) = 90\pi$. M1: two circular ends $= 2 \times \pi(3)^2 = 18\pi$. M1: total $= 108\pi$. A1: 339 cm^2 (3 s.f.). [4]
18(a)	200 g	B1: when $t = 0$, $M = 200 \times 1 = 200$. [1]
18(b)	107.9 g (1 d.p.)	M1: $M = 200 \times (0.94)^{10}$. A1: 107.9 g. [2]
18(c)	11 years	M1: sets up $200 \times (0.94)^t = 100 \Rightarrow (0.94)^t = 0.5$. M1: trial $t = 11$: $200 \times 0.94^{11} \approx 101.4$. Trial $t = 12$: ≈ 95.3 . A1: half-life ≈ 11 years (accept 11 or 12). [3]
19(a)	Angle AOB = 64°	B1: 64° . B1: reason "angle at centre is twice angle at circumference". [2]
19(b)	Angle ADB = 148°	M1: ACBD is a cyclic quadrilateral; angle ACB + angle ADB = 180° . A1: angle ADB = $180 - 32 = 148^\circ$. (Reason: opposite angles of cyclic quadrilateral sum to 180° .) [2]
20(a)	Shown	M1: $x^3 + 2x - 7 = 0 \Rightarrow x^3 = 7 - 2x$. A1: takes cube root: $x = (7 - 2x)^{1/3}$. [2]
20(b)	$x_1 = 1.5874$, $x_2 = 1.5733$, $x_3 = 1.5757$	M1: $x_1 = (7 - 3)^{1/3} = 4^{1/3} = 1.5874$. M1: applies iteration twice more. A1: all three values correct to 4 d.p. [3]
21(a)	$x = 5$	M1: $5 + 8 + 7 + x + 3 + 2 = 30$. A1: $x = 5$. [2]
21(b)	Mean = 1.97	M1: $\Sigma fx = 0(5) + 1(8) + 2(7) + 3(5) + 4(3) + 5(2) = 59$. M1: divides by 30. A1: $59/30 = 1.97$ (2 d.p.). [3]
22(a)	20 minutes	B1: identifies the horizontal section from 20 to 40 min. [1]
22(b)	16 km/h	M1: total distance $= 12 + 12 = 24$ km. M1: total time $= 90 \text{ min} = 1.5$ hours. A1: $24/1.5 = 16 \text{ km/h}$. [3]
22(c)	Return journey (60–90 min); speed = 24 km/h	M1: calculates speed for each section: 0–20 min: 8 km in 20 min $= 24 \text{ km/h}$. 40–60 min: 4 km in 20 min $= 12 \text{ km/h}$. 60–90 min: 12 km in 30 min $= 24 \text{ km/h}$. M1: identifies fastest sections (tied at 24 km/h). A1: states 0–20 min and 60–90 min are tied at 24 km/h. [3]

Worked Solutions — Predicted Paper 3 Set B

Higher Tier · Edexcel 1MA1 style

Q1: Successive percentages

First reduction: $85 \times (1 - 0.35) = 85 \times 0.65 = £55.25$.

Second reduction: $55.25 \times (1 - 0.20) = 55.25 \times 0.80 = \textbf{£44.20}$.

Q2: Standard form mixed

$$(4.5 \times 10^7) \div (1.5 \times 10^2) = 3 \times 10^5.$$

$$\text{Add } 2 \times 10^5: \text{result} = (3 + 2) \times 10^5 = \textbf{5} \times \textbf{10}^5.$$

Q3: Ratio (equalising)

Total parts = $5 + 7 = 12$. For equal split, each must have 6 parts.

Kate must give 1 part (going from 7 to 6 parts) to Jenny (from 5 to 6).

So 1 part = £36. Total sum = $12 \times 36 = \textbf{£432}$.

(Check: Jenny had $5 \times 36 = £180$, Kate had $7 \times 36 = £252$. After transfer: both have £216. ✓)

Q4: Linear equation with fractions

Multiply both sides by 12 (LCM of 4 and 3):

$$12 \times [(x+3)/4 + (2x-1)/3] = 12 \times 5.$$

$$3(x + 3) + 4(2x - 1) = 60.$$

$$3x + 9 + 8x - 4 = 60 \Rightarrow 11x + 5 = 60.$$

$$11x = 55 \Rightarrow \textbf{x = 5}.$$

Q5: Finding rate from compound interest

$$\text{Let } r \text{ be the rate. } 2500 \times (1 + r)^2 = 2652.25.$$

$$(1 + r)^2 = 2652.25 / 2500 = 1.0609.$$

$$1 + r = \sqrt{1.0609} = 1.03.$$

$$r = 0.03 = \textbf{3\% per year}.$$

Q6: Bearings + Pythagoras

(a) The boat sails from A on bearing 060° , so the direction AB points 60° clockwise from north.

At B, the reverse direction (B to A) is $060 + 180 = 240^\circ$.

The new direction BC is 150° .

$$\text{Interior angle } ABC = 240 - 150 = \textbf{90}^\circ. \checkmark$$

(b) Right triangle ABC with legs 5 km and 4 km.

$$AC = \sqrt{(25 + 16)} = \sqrt{41} = \textbf{6.40 km (3 s.f.)}.$$

(c) $\tan(\text{angle BAC}) = \text{opposite/adjacent} = 4/5 = 0.8$.

$$\text{Angle BAC} = \tan^{-1}(0.8) = 38.66^\circ.$$

$$\text{Bearing of C from A} = 060 + 38.66 \approx \textbf{099}^\circ \text{ (nearest degree)}.$$

Q7: Surds

(a) $\sqrt{72} = \sqrt{(36 \times 2)} = 6\sqrt{2}$. $\sqrt{50} = \sqrt{(25 \times 2)} = 5\sqrt{2}$.

Sum = $11\sqrt{2}$.

(b) $8\sqrt{2} = 8\sqrt{2} \times \sqrt{2}/\sqrt{2} = 8\sqrt{2}/2 = 4\sqrt{2}$.

Q8: Inverse proportion

Total work = 4 builders $\times$ 12 days = 48 builder-days.

With 6 builders: time = $48 / 6 = 8$ days.

Q9: Cumulative frequency

(a) Median at cf = 50. Reading across: $m \approx 47$ kg.

(b) Q1 at cf = 25: $m \approx 35$ kg. Q3 at cf = 75: $m \approx 53$ kg.

IQR $\approx 53 - 35 = 18$ kg.

(c) At $m = 65$, cf ≈ 84 . Number above = $100 - 84 = 16$ packages.

Q10: Completing the square

$$x^2 - 6x + 2 = 0.$$

$$(x - 3)^2 - 9 + 2 = 0.$$

$$(x - 3)^2 = 7.$$

$$x - 3 = \pm\sqrt{7}.$$

$$x = 3 \pm \sqrt{7} \text{ (so } a = 3, b = 1, c = 7\text{)}.$$

Q11: Probability with conditional

(a) $P(\text{both black}) = (5/9) \times (4/8) = 20/72$.

$$P(\text{both white}) = (4/9) \times (3/8) = 12/72.$$

$$P(\text{same colour}) = 20/72 + 12/72 = 32/72 = 4/9.$$

(b) $P(\text{both black} \mid \text{same colour}) = P(\text{both black}) / P(\text{same colour})$.

$$= (20/72) / (32/72) = 20/32 = 5/8.$$

Q12: Trapezium

(a) Area = $(1/2)(a + b)h = (1/2)(8 + 14)(6) = 66 \text{ cm}^2$.

(b) Drop perpendicular from B to DC, meeting at E. Then BE = AD = 6 cm.

DE = AB = 8 cm (since AB $\parallel$ DC and AD $\perp$ DC). So EC = $14 - 8 = 6$ cm.

Right triangle BEC: $BC^2 = 6^2 + 6^2 = 72$.

$$BC = \sqrt{72} = 6\sqrt{2} \approx 8.49 \text{ cm (3 s.f.)}.$$

Q13: Perpendicular bisector

Midpoint of (1, 4) and (5, -2) = $((1+5)/2, (4-2)/2) = (3, 1)$.

Gradient of original line = $(-2 - 4) / (5 - 1) = -6/4 = -3/2$.

Perpendicular gradient = $2/3$.

$$\text{Equation: } y - 1 = (2/3)(x - 3) \Rightarrow y = (2/3)x - 2 + 1.$$

$$y = (2/3)x - 1.$$

Q14: Vectors

(a) $\mathbf{AB} = \mathbf{OB} - \mathbf{OA} = \mathbf{b} - \mathbf{a}.$

$$\mathbf{AX} = (1/4)\mathbf{AB} = (1/4)(\mathbf{b} - \mathbf{a}).$$

$$\mathbf{OX} = \mathbf{OA} + \mathbf{AX} = \mathbf{a} + (1/4)(\mathbf{b} - \mathbf{a}) = (3/4)\mathbf{a} + (1/4)\mathbf{b}.$$

(b) $\mathbf{OY} = (1/4)\mathbf{b}.$

$$\mathbf{XY} = \mathbf{OY} - \mathbf{OX} = (1/4)\mathbf{b} - (3/4)\mathbf{a} - (1/4)\mathbf{b} = -(3/4)\mathbf{a}.$$

Since $\mathbf{XY}$ is a scalar multiple of $\mathbf{a}$ ($= \mathbf{OA}$), $\mathbf{XY}$ is parallel to $\mathbf{OA}$. ✓

Q15: Algebraic proof

$$(n+2)^2 - (n-2)^2 = (n^2 + 4n + 4) - (n^2 - 4n + 4).$$

$$= n^2 + 4n + 4 - n^2 + 4n - 4.$$

$$= 8n.$$

Since n is any integer, $8n$ is always a multiple of 8. ✓

Q16: Rearranging formula

$$y(2 - x) = 3x + 1.$$

$$2y - xy = 3x + 1.$$

Move x terms to one side: $2y - 1 = 3x + xy = x(3 + y).$

$$x = (2y - 1) / (y + 3).$$

Q17: Cylinder

(a) Volume = $\pi r^2 h = \pi(3)^2(15) = 135\pi \text{ cm}^3 (\approx 424 \text{ cm}^3).$

(b) Curved surface area = $2\pi r h = 2\pi(3)(15) = 90\pi.$

$$\text{Two circular ends} = 2 \times \pi(3)^2 = 18\pi.$$

$$\text{Total surface area} = 90\pi + 18\pi = 108\pi \approx 339 \text{ cm}^2 \text{ (3 s.f.)}.$$

Q18: Exponential decay

(a) When $t = 0$: $M = 200 \times 1 = 200 \text{ g}.$

(b) $M = 200 \times (0.94)^{10} = 200 \times 0.53862 = 107.7 \text{ g (1 d.p.)}.$

(Slight variation: $200 \times 0.94^{10} \approx 107.7\text{--}107.9$ depending on calculator precision.)

(c) Set $200 \times (0.94)^t = 100$, so $(0.94)^t = 0.5.$

$$\text{Trial: } t = 11 \Rightarrow (0.94)^{11} \approx 0.507; t = 12 \Rightarrow \approx 0.476.$$

So half-life is **about 11 years** (closest whole-year value).

Q19: Circle theorems

(a) By the 'angle at centre is twice angle at circumference' theorem:

$$\text{Angle AOB} = 2 \times \text{angle ACB} = 2 \times 32 = 64^\circ.$$

(b) ACBD is a cyclic quadrilateral (all four points lie on the circle).

$$\text{Opposite angles sum to } 180^\circ: \text{angle ACB} + \text{angle ADB} = 180.$$

Angle ADB = $180 - 32 = 148^\circ$.

Q20: Iteration

(a) Start: $x^3 + 2x - 7 = 0 \Rightarrow x^3 = 7 - 2x$.

Take cube root: $x = (7 - 2x)^{1/3}$. ✓

(b) Starting with $x_0 = 1.5$:

$$x_1 = (7 - 3)^{1/3} = 4^{1/3} = \mathbf{1.5874}.$$

$$x_2 = (7 - 2(1.5874))^{1/3} = (3.8253)^{1/3} = \mathbf{1.5640}.$$

$$x_3 = (7 - 2(1.5640))^{1/3} = (3.8720)^{1/3} = \mathbf{1.5703}.$$

(Values may vary slightly with rounding; accept ± 0.001 .)

Q21: Frequency table with unknown

(a) $5 + 8 + 7 + x + 3 + 2 = 30$. So $25 + x = 30 \Rightarrow \mathbf{x = 5}$.

(b) $\Sigma fx = 0(5) + 1(8) + 2(7) + 3(5) + 4(3) + 5(2)$

$$= 0 + 8 + 14 + 15 + 12 + 10 = 59.$$

$$\text{Mean} = 59 / 30 \approx \mathbf{1.97 \text{ goals per match (2 d.p.)}}.$$

Q22: Travel graph

(a) Jamie stopped during the flat section from 20 to 40 min, i.e. for **20 minutes**.

(b) Total distance = 12 km out + 12 km back = 24 km. Total time = 90 min = 1.5 hours.

$$\text{Average speed} = 24 / 1.5 = \mathbf{16 \text{ km/h}}.$$

(c) Calculate speed per section (using distance and time in hours):

$$0\text{--}20 \text{ min: } 8 \text{ km in } 1/3 \text{ h} \Rightarrow 24 \text{ km/h}.$$

$$40\text{--}60 \text{ min: } 4 \text{ km in } 1/3 \text{ h} \Rightarrow 12 \text{ km/h}.$$

$$60\text{--}90 \text{ min: } 12 \text{ km in } 0.5 \text{ h} \Rightarrow 24 \text{ km/h}.$$

Fastest sections: **0–20 min and 60–90 min (tied at 24 km/h)**.